Near-side azimuthal and pseudo-rapidity correlations of neutral strange baryons and mesons in d+Au and Au+Au collisions at $\sqrt{s_{NN}} = 200$ GeV

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We present results on near-side di-hadron correlations of neutral strange baryons (Λ,$\bar{\Lambda}$) and mesons (K_S^0) in the intermediate p_T regime (p_T = 2-6 GeV/c) in d+Au and Au+Au collisions at $\sqrt{s_{NN}} = 200$ GeV measured by the STAR experiment at RHIC. The study is performed for both identified strange trigger particles associated with charged particles as well as charged trigger particles using strange associated particles. The di-hadron distribution in ($\Delta\phi$, $\Delta\eta$) space is decomposed into a narrow jet-like peak at small angular separation $\Delta\phi$ and a component which is narrow in $\Delta\phi$ and extended in $\Delta\eta$, the "ridge". We study the near-side yield of the ridge and jet-like components as a function of centrality of the collision and transverse momentum (p_T) of trigger and associated particles to look for possible flavor and baryon/meson differences. The magnitude, p_T spectra and particle composition of the jet-like component are similar to those in d+Au collisions. The strength of the ridge increases steeply with centrality in Au+Au collisions for all studied particle species. The p_T spectra and baryon/meson composition of the ridge resemble those for the bulk particle production. The results on the ridge are discussed with theoretical models.

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I. INTRODUCTION

Ultra-relativistic heavy-ion collisions create a unique environment for the investigation of nuclear matter at extreme temperatures and energy densities in the laboratory. Already the first measurements of nuclear modification factors [1, 2] and anisotropic elliptic flow at the Relativistic Heavy Ion Collider (RHIC) at $\sqrt{s_{NN}} = 200$ GeV showed that the nuclear medium created is opaque to particles with large transverse momentum (p_T), is described by partonic degrees of freedom and has properties close to those expected of a perfect fluid.

As compared to nuclear modification factors, studies of jets and their modification in the medium provide a better sensitivity to the medium properties [3]. Azimuthal correlations of particles with large p_T can be used to study processes associated with jets. Previous investigations of two-particle azimuthal correlations in central Au+Au collisions span a large p_T range (p_T = 0.15 - 15 GeV/c) resulted in several interesting observations. There are striking differences between those correlations in p+p and d+Au and those in central Au+Au collisions: a) the disappearance of the away-side jet at intermediate p_T = 2-6 GeV/c consistent with partonic energy loss in medium [4], b) the increase of the number of low p_T (p_T = 0.15-2 GeV/c) particles on the away-side from the trigger particle with concomitant shape modifications suggesting conical particle emission [5], c) *Punch through* or the ability of away-side jets, with sufficiently large p_T ($p_T > 8$ GeV/c), to traverse the hot and dense medium [6], and d) the presence of a correlation on the near-side extended in pseudo-rapidity ($\Delta\eta$), commonly referred to as the *ridge* at intermediate p_T [7, 8].

It is interesting to note that besides the shape modifications of jet-like correlations and the existence of the ridge in the intermediate transverse momentum range, the production mechanism of hadrons seems to differ from simple fragmentation. Baryon production is less suppressed in central Au+Au collisions than that of mesons resulting in baryon/meson ratios enhanced in central Au+Au collisions relative to p+p and d+Au collisions [9, 10]. The measured baryon/meson ratios increase with p_T until reaching their maximum value of ≈ 3 relative to p+p colliisions at $p_T \approx 3$ GeV/c in both the strange and non-strange quark sectors. A fall-off of the baryon/meson ratio is observed for $p_T > 3$ GeV/c and both the strange and non-strange baryon/meson ratios in Au+Au collisions approach the values measured in p+p collisions at $p_T \approx 6$ GeV/c.

This shows that in the intermediate p_T range at RHIC energies unmodified jet fragmentation is not the dominant source of particle production. Parton recombination and coalescence models are an alternative particle production mechanism [11–15]. In these models, competition between recombination and fragmentation results in a shift in the onset of the perturbative regime to higher transverse momenta, $p_T \approx$ 4-6 GeV/c. Due to the steeply falling transverse momentum spectrum of partons, fragmentation is a much less efficient particle production mechanism than recombination. Moreover, the steeply falling parton spectrum favors the recombination of three quarks to form a baryon over the recombination of two quarks to form a meson with the same p_T. Such mechanisms naturally lead to increased production of baryons relative to mesons, in qualitative agreement with the data.

In this paper, studies of two-particle correlations on the near-side in d+Au and Au+Au collisions at $\sqrt{s_{NN}} = 200$ GeV measured by the STAR experiment are presented. In particular, results on two-particle correlations in pseudo-rapidity and azimuth for charged particles, neutral strange baryons (Λ,$\bar{\Lambda}$) and mesons (K_S^0) in the in-

TABLE I: Selection criteria applied for K_S^0, Λ and $\bar{\Lambda}$ particle identification. DCA denotes the Distance of Closest Approach. [ADD LAMBDA MASS CUT]

Cut	K_S^0	Λ	$\overline{\Lambda}$
DCA of V^0 to primary vertex	< 0.6 cm	< 0.6 cm	< 0.6 cm
DCA of positive daughter to primary vertex	> 1.2 cm	> 0.7 cm	> 1.2 cm
DCA of negative daughter to primary vertex	> 1.2 cm	> 1.2 cm	> 0.7 cm
DCA of daughters at decay vertex	< 0.8 cm	< 0.8 cm	< 0.8 cm
N(TPC hits) daughters	> 20	> 20	> 20
$\mathrm{N}(\sigma)(dE/dx)$	< 3	< 3	< 3
V^0 decay length	> 12 cm	> 12 cm	> 12 cm

TABLE II: Definition of centrality bins. Each centrality bin is defined by the fraction of the geometric cross-section, the primary charged tracked track multiplicity N_{ch} in the TPC for $|\eta| < 0.5$, the average number of participating nucleons $\langle N_{part} \rangle$ and the average number of inelastic binary nucleon-nucleon collisions $\langle N_{bin} \rangle$.

% of cross-section	N_{ch}	$\langle N_{part} \rangle$	$\langle N_{bin} \rangle$
0-12%	N/A	$315.7^{+5.6}_{-4.5}$	900^{+71}_{-64}
0-10%	> 441	$325.9^{+5.4}_{-5.3}$	940^{+67}_{-70}
10-20%	319-441	$234.6^{+8.3}_{-9.3}$	591^{+52}_{-60}
20-30%	222-319	$166.7^{+9.0}_{-11}$	369^{+41}_{-51}
30-40%	150-222	$115.5^{+8.7}_{-11}$	220^{+30}_{-38}
40-80%	14-150	$41.5^{+7.0}_{-6.6}$	57^{+14}_{-13}

termediate p_T (p_T = 2-6 GeV/c) regime are discussed. Both identified strange trigger particles associated with charged particles and charged trigger particles using identified strange associated particles are studied. The near-side yield is studied separately for the jet-like and ridge components as a function of centrality of the collision, pseudo-rapidity and transverse momentum of trigger and associated particles to look for possible flavor, baryon/meson and particle/antiparticle differences. We compare our results to several theoretical models attempting to describe the origin of the ridge.

II. EXPERIMENTAL SETUP AND PARTICLE RECONSTRUCTION

The data presented in this paper were measured by the STAR experiment at $\sqrt{s_{NN}}$ = 200 GeV at RHIC. The data sample consists of 12 million of d+Au collisions taken in 2003, and 25 million minimally biased and 24 million central triggered Au+Au events corresponding to the 12% most central collisions taken in 2004. STAR [16] is a large acceptance, multi-purpose spectrometer consisting of several detectors inside a large solenoidal magnet with a uniform magnetic field of 0.5 T applied along the beam line. The analysis presented here is based exclusively on charged particle tracks detected and reconstructed in the Time Projection Chamber (TPC) [17]. The TPC has a pseudo-rapidity acceptance $|\eta| < 1.5$ and full azimuthal coverage, and is thus well suited for correlation studies. The TPC has 45 pad rows in the radial direction allowing up to 45 independent spatial and energy loss (dE/dx) measurements along each charged particle track. The momentum resolution is determined to be $\Delta k/k \sim 0.0078 + 0.0098 \cdot p_T$ (GeV/c) [17]. We identify weakly decaying neutral strange (V^0) particles Λ, $\bar{\Lambda}$ and K_S^0 by topological reconstruction of their decay vertices from their charged hadron daughters measured in the TPC [18]:

$$\begin{aligned} \Lambda &\rightarrow p + \pi^-, BR = (63.9 \pm 0.5)\% \\ \bar{\Lambda} &\rightarrow \bar{p} + \pi^+, BR = (63.9 \pm 0.5)\% \\ K_S^0 &\rightarrow \pi^+ + \pi^-, BR = (68.95 \pm 0.14)\% \end{aligned} \quad (1)$$

where BR denotes the branching ratio.

The STAR V^0 finding reconstruction software pairs oppositely charged particle tracks into V^0 candidates using rather loose cuts. In order to obtain a very clean sample of strange particles, additional, more stringent topological cuts were applied. [C: Too many details? Cut. Or need to add d+Au data, which has different cuts and change text to just list the final cuts. Intermediate cuts irrelevant.] The selection cuts applied for $p_T > 2$ GeV/c, which yield signal to background ratio of $\geq$ 15:1, are summarized in Table I.

In order to determine the reconstruction efficiency of particles in the TPC, a Monte-Carlo simulation was performed. In this simulation, a Monte-Carlo generator is used to produce particles of interest through the GEANT simulation of the STAR detector with a flat transverse momentum distribution. The resulting signal (at the level of the detector response) is then embedded into real events to assure a realistic track finding environment. The multiplicity of embedded particles is 5% of the total charged particle event multiplicity. After processing these embedded events with the same reconstruction software used to reconstruct the data, the efficiency correction is determined by comparing the input and reconstructed tracks.

For charged particles with $p_T > 1.0$ GeV/c and $|\eta| < 1.0$, which are of the central interest in this paper, the TPC charged tracking efficiency in d+Au collisions is 89% and varies from 70% in the most central to 84% in the peripheral Au+Au events at $\sqrt{s_{NN}}$ = 200 GeV. The V^0 reconstruction efficiency is lower than that of charged particles because it is a convolution of two charged particle reconstruction efficiencies, finite geometrical acceptance, the V^0-finding software and V^0 selection criteria applied. Typical values of V^0 reconstruction efficiency in central Au+Au collisions are 10-15%.

We analyzed events from a centrality interval corresponding to 0-80% of the hadronic interaction cross-section with a primary vertex within 30 cm of the TPC center along the beam direction. We define the centrality of an event from the number of charged tracks in the TPC with $|\eta| < 0.5$, $p_T > 0.2$ GeV/c, more than 10 measured space points in the TPC and distance of closest approach to the primary vertex (DCA) less than 2 cm, as commonly used in previous STAR analyses. The definition of the centrality bins is given in Table II, where each centrality bin is defined by the fraction of the geometric hadronic cross-section, the primary charged tracked track multiplicity N_{ch} in the TPC for $|\eta| < 0.5$, the average number of participating nucleons $\langle N_{part}\rangle$ and the average number of inelastic binary nucleon-nucleon collisions $\langle N_{bin}\rangle$.

III. DATA ANALYSIS

A. Two-particle correlation method

Below we introduce the two-particle correlation method and define particle yields which will be used throughout the rest of the paper. First, events with at least one particle with high transverse momentum, which we refer to as the *trigger particle*, are selected. The transverse momentum of the trigger particle, p_T^{trig}, has to satisfy the condition $p_T^{trig}(min) < p_T^{trig} < p_T^{trig}(max)$. Second, for each trigger particle in a given event we select all *associated particles* with transverse momentum p_T^{assoc} satisfying the requirement $p_T^{assoc}(min) < p_T^{assoc} < p_T^{trig}$. The azimuthal angular difference $\Delta\phi = \phi^{trig} - \phi^{assoc}$ and pseudo-rapidity difference $\Delta\eta = \eta^{trig} - \eta^{assoc}$ between the trigger and associated particles are calculated and incremented to the two-particle distribution function defined as:

$$C(\Delta\phi, \Delta\eta) = \frac{1}{N_{trig}} \iint dp_T^{trig} dp_T^{assoc} \frac{dN_{pair}(p_T^{trig}, p_T^{assoc}, \Delta\phi, \Delta\eta)}{\epsilon(p_T^{assoc}) acc(\Delta\phi, \Delta\eta)} \tag{2}$$

where N_{trig} is the number of trigger particles, ϵ is the reconstruction efficiency of associated particles, and $acc(\Delta\phi, \Delta\eta)$ is the correction for the finite pair acceptance. The pair acceptance correction is computed using a mixed event technique. Azimuthal inefficiencies due to the TPC sector boundaries are reflected in the acceptance correction as small gaps in $\Delta\phi$. The pair acceptance correction in $\Delta\eta$ has a triangular shape resulting from a convolution of two uniform single track η acceptances.

In this paper we concentrate on the study of the near-side correlation peak. To be consistent with previous STAR analyses, we project the two-dimensional distribution from Eq.(2) onto $\Delta\phi$ in the interval $|\Delta\eta| < 1.7$. The distribution around the near-side peak is fit with a Gaussian sitting on top of a flat background in d+Au and an elliptic flow (v_2) modulated background in Au+Au collisions, respectively. To adjust the level of the elliptic flow modulated background we use the ZYAM (=Zero Yield At Minimum) method [19]. In order to avoid problems which could arise from statistical fluctuations, the level of the v_2 modulated background is determined as an average from several (typically 3 to 5) neighbouring $\Delta\phi$ bins in the "valley region", $\Delta\phi \approx 1$.

The near-side yield of associated particles is eventually calculated as the area under the Gaussian peak. This corresponds to the average number of particles correlated with the trigger particle within the kinematic cuts. We study the jet-like and ridge contributions to the near-side yield separately by analyzing the correlations in two $\Delta\eta$ windows: $|\Delta\eta| < 0.7$ containing both jet-like and ridge correlations, and $|\Delta\eta| > 0.7$ containing only the ridge contributions, assuming the jet-like contribution at large $\Delta\eta$ is negligible. Assuming the uniformity of v_2 with η within the TPC acceptance [20], the elliptic flow contribution cancels out. Therefore there is no systematic error on the jet-like yield due to the error on v_2.

IV. RESULTS

A. V^0-charged correlations

In this section we discuss properties of the near-side azimuthal correlations using identified strange trigger particles (Λ, $\bar{\Lambda}$, and K_S^0) and unidentified charged particles associated with charged particles. Comparing results from

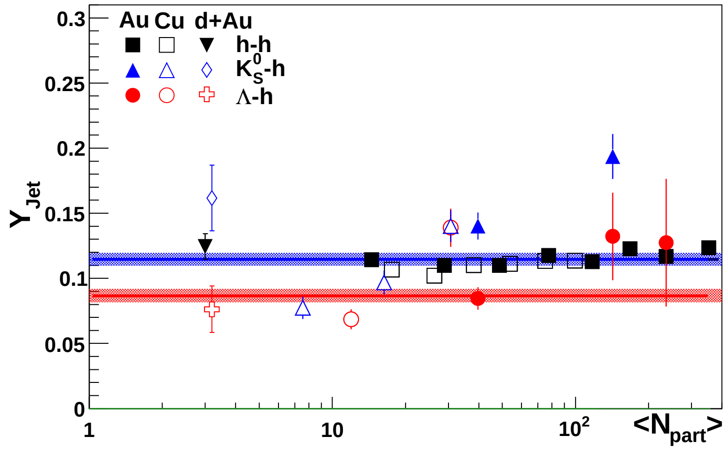

FIG. 1: Centrality dependence of the jet-like yield of associated charged particles for various trigger species in d+Au, Cu+Cu and Au+Au collisions.

d+Au and various centralities in Au+Au collisions at $\sqrt{s_{NN}}$ = 200 GeV, we observe an increase of the near-side yields by a factor of 3-4 going from d+Au to central Au+Au collisions for all studied trigger species. Figure 1 shows the centrality dependence of the ridge and jet-like yields of the near-side correlation. While the jet-like yield is approximately independent of centrality and consistent with its value in d+Au, the ridge yield increases steeply with centrality.

The jet-like yield is free of systematic errors due to the elliptic flow subtraction (cf. Section III A), however the ridge yield is not. The systematic errors due to the v_2 subtraction are shown in Figure 1 left as bands around the data points. These systematic errors were estimated by subtracting the v_2 measured by the event plane method (the lower bound) and by the 4-particle cumulant method (the upper bound) [21]. The uncertainties in the elliptic flow subtraction result in a systematic error of roughly 30% on the extracted associated yield and are point-to-point correlated.

Next, we study the dependence of ridge and jet-like correlations on transverse momentum of the trigger particle, p_T^{trig}. Figure **??** shows the p_T^{trig} dependence of Gaussian widths of jet-like peaks measured independently from $\Delta\eta$ and $\Delta\phi$ projections. The Gaussian width in $\Delta\eta$ decreases steeply with increasing p_T^{trig}. The evolution of the jet-like peak width in $\Delta\phi$ is more gradual. For $p_T^{trig} > 3$ GeV/c, both widths are comparable. We do not observe a clear dependence among trigger particle species studied. The dependence of the extracted jet-like and ridge yields on the transverse momentum of the trigger particle, p_T^{trig}, is shown in Figure 2. While the ridge yield increases with p_T^{trig} and flattens off for $p_T^{trig} > 3$ GeV/c, the jet-like yield monotonically increases with p_T^{trig} for $p_T^{trig} > 3$ GeV/c as expected for jet-like processes. The jet-like yield for Λ triggers is systematically below that of charged hadron and K^0_S triggers. The effects of artificial track merging, which result in the yield loss, affect Λ triggered correlations more than those where charged tracks or K^0_S are used as trigger particles. This effect is about 15-20% and has been in detail discussed in Section IIIB and Figure **??**. The systematic errors due to the v_2 subtraction have been calculated using the same procedure as in Figure 1 and are again shown as bands around the data points.

We also measured the invariant p_T spectra of associated charged particles for p_T^{trig} = 3-6 GeV/c in central Au+Au collisions as shown in Figure 3. We fit the distributions with an exponential function $e^{-p_T/T}$ and the inverse slope, T, is for each trigger species given in Table III. The ridge spectra have the inverse slope of about T = 400-440 MeV, close to that of the bulk when the fit is performed in the same p_T range. The inverse slope of the jet-like spectra is by few tens of MeV larger than that of ridge spectra. Those studies demonstrate that the inverse slope values are within errors independent of trigger particle species. A detailed investigation of the evolution of the inverse slope of the jet-like and ridge spectra with p_T^{trig} has been carried out with charged trigger and associated particles [7, 8], where the statistics are richer than for identified correlations. A clear difference in the evolution of the slope with p_T^{trig} was observed for jet-like and ridge-like correlations. While the inverse slope of the ridge spectra remains constant with increasing p_T^{trig} and differs from the bulk by about 50 MeV, the jet-like spectra show a steep increase of the slope with p_T^{trig}.

B. Charged-V^0 correlations

It is also important to study the composition of particles in the ridge and jet-like components in order to look for a possible enhancement of the baryon/meson ratio which is present in the inclusive p_T distributions in Au+Au collisions at RHIC. For this purpose, we have

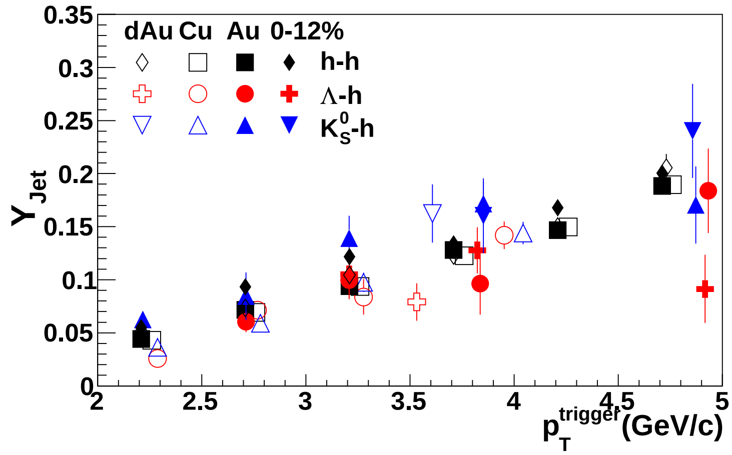

FIG. 2: Dependence of the ridge yield (a) and jet-like yield (b) on p_T^{trig} for various trigger species in central (0-10%) Au+Au collisions. The bands indicate systematic errors on the ridge yield due to the v_2 subtraction.

studied two-particle correlations using charged trigger particles with which were associated with Λ and K^0_S particles. Using the same method as above, we extracted the jet-like and ridge yields which were then used to calculate the Λ/K^0_S ratio in the jet-like and ridge components. The results obtained are shown in Figure 4 together with the ratio measured from the inclusive p_T spectra [22]. The Λ/K^0_S ratio calculated for the jet-like component is 0.46±0.21 and consistent with that measured in p+p. The same ratio in the ridge is 0.81±0.14, higher than in the jet-like component. The large statistical errors do not allow to draw a definite conclusion on the origin of the ridge yet... Jana: working on a new figure with improved topological cuts to preserve as much signal as possible, ultimately we will need L2gamma triggered Run7 data, not for this paper ...

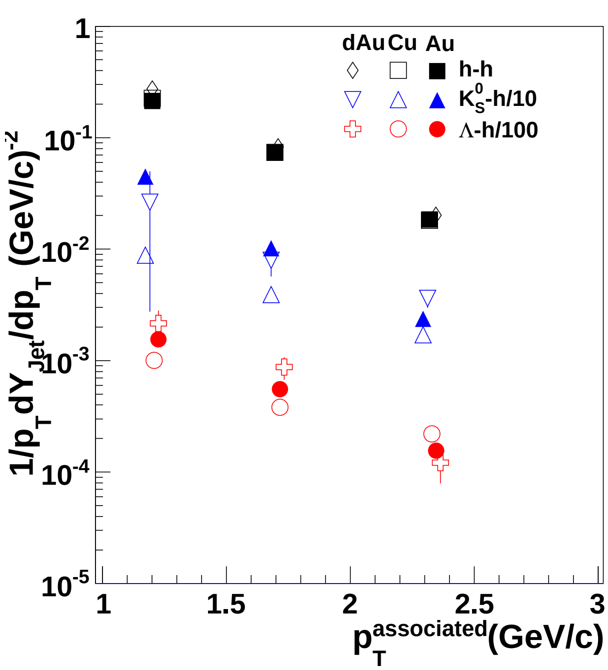

FIG. 3: Invariant p_T distribution of associated charged particles in the jet-like peak and ridge in central Au+Au collisions for various trigger species indicated by the legend. The lines are exponential fits to the data.

V. SUMMARY AND DISCUSSION

We have reported results on the properties of the near-side correlation peak at intermediate-p_T at RHIC using strange particles and unidentified charged hadrons. The measured correlations reveal a strong contribution from long-range $\Delta\eta$ correlations on the near-side for all particle species studied.

The strength of these ridge-like correlations increases steeply with centrality. The inverse slope of the associated particle spectra in the ridge is about 50 MeV higher that that of particles in the bulk. Better statistics are needed to draw final conclusions on the baryon/meson ratios in the ridge, but the trend seems to confirm that the ridge ratios are in better agreement with bulk production than with jet fragmentation. This is also confirmed by preliminary STAR results for the p/π ratios [23]. The jet-like particle yield is independent of centrality in Au+Au collisions within errors and consistent with the yield in d+Au collisions, within error. In addition the baryon/meson ratios in the jet-like component in Au+Au are close to those in p+p collisions as shown in Figure 4. While the jet-like correlations are in agreement with a simple fragmentation process, the origin of the ridge is not well understood.

This phenomenon has triggered significant interest in the theoretical community and several models are currently available to describe the observed ridge. Below we discuss their main features.

A model based on the recombination and fragmentation of partons in the medium [24], links the origin of the ridge to the longitudinal expansion of the thermal partons that are enhanced by the energy loss of a hard parton traversing the medium. The model predicts that the baryon/meson ratio of particles associated with the ridge should follow that of particles produced in the bulk and therefore lead to an increased production of baryons. Our data support this prediction, although the errors are large. In addition, the model predicts the inverse slope of the enhanced thermal parton distribution to be larger than that of particles in the bulk by only 15 MeV. The data show that this difference is ≈ 50 MeV over a broad range of $p_T^{trig} = 3\text{-}12$ GeV/c.

In another approach, the interaction of high-p_T partons with a dense medium in the presence of strong longitudinal collective flow is predicted to lead to a characteristic breaking of the rotational symmetry of the average jet energy and multiplicity distribution in the $\eta \times \phi$ plane [25]. This will in turn cause a medium-induced broadening of gluon radiation in pseudo-rapidity and form a ridge in $\Delta\eta$.

The mechanism proposed in [26] relates the origin of the ridge to the longitudinal expansion of the medium and spontaneous formation of extended color fields in such expanding medium due to the presence of plasma instabilities. The predicted ridge at low transverse momenta is expected to decrease with increasing energy of the associated radiation. The momentum range of the partons contained in the ridge is in the recombination regime and therefore the authors of this model predict that it would reflect itself in the baryon to meson ratio of associated hadrons.

Another suggested mechanism for the origin of the ridge is based on jet quenching combined with strong radial flow [27, 28]. The radial expansion of the system creates strong position-momentum correlations leading to characteristic rapidity, azimuthal and p_T correlations among produced particles.

In the momentum kick model [29], partons in the medium suffer a collision with a propagating jet and acquire in such collision a momentum kick along the jet direction forming a ridge structure.

Finally a recent paper based on the radial expansion of particle originating from a Color Glass Condensate (CGC) flux tube is able to describe the centrality dependence and system size dependence of the ridge, when normalized to the ridge strength in central Au+Au collisions [30].

TABLE III: The inverse slope of charged particle associated p_T spectra for the ridge (T_{ridge}) and the jet-like ($T_{jet-like}$) componenents in central Au+Au collisions at $\sqrt{s_{NN}} = 200$ GeV. The errors quoted are statistical only.

Trigger particle	T_{ridge} [MeV]	$T_{jet-like}$ [MeV]
$h^{\pm}$	438±4	478±8
K_S^0	406±20	530±61
Λ	416±11	445±49

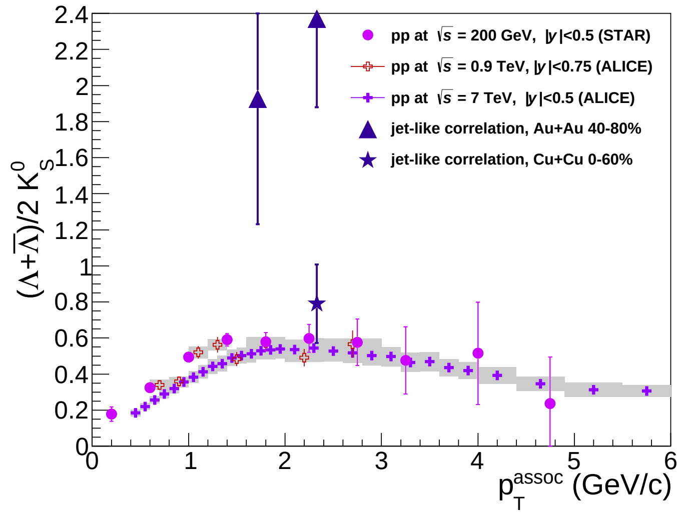

FIG. 4: Λ/K_S^0 ratio measured in inclusive p_T distributions, near-side jet-like and ridge-like correlation peaks in Au+Au collisions together with this ratio obtained from inclusive p_T spectra in p+p collisions.

In order to draw final conclusions about the origin of the ridge, more actual theoretical calculations are needed. No model has made quantitative predictions on the strength, particle distribution and kinematics of the ridge. This is a very active field though and we look forward to theoretical progress in order to understand our data. On the experimental side, the particle identified distributions in the jet-like and ridge components are limited by statistics, and the new, high statistics Au+Au data at RHIC will yield more conclusive evidence for potential differences in the hadron production mechanisms in correlated structures in relativistic heavy ion collisions.

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